

Execution Plan — Nano-Rare Gene Therapy Matching Framework

Overview

This plan decomposes the project into 6 milestones across 12 weeks, with a concentrated 2-week MVP sprint for Week 1-2. The architecture is modular Python (CLI → local analysis → reports), with each module independently testable.

Milestones

Milestone	Week	Deliverable
M1 — PoC Pipeline	1-2	Single-disease end-to-end match via CLI
M1b — ROGDI Deep-Dive	2-3	Full 9-dimension analysis + Standardised GT Protocol for ROGDI
M2 — Multi-Disease Batch	3-4	Batch query Orphanet + OMIM, ranked output
M3 — Full Scoring Matrix	5-6	All 9 dimensions implemented + weighted aggregation
M4 — Regulatory Brief	7-8	Auto-generated Standardised Gene Therapy Protocol (ROGDI first)
M5 — Validation	9-10	Retrospective validation case studies (ROGDI illustrative)
M6 — Public Release	11-12	Open-source publication, documentation, preprint with ROGDI results

M1: PoC Pipeline (Week 1-2)

Week 1: Foundation + Disease Query Module

Monday: Project Setup + Architecture - Create Python project structure with `pyproject.toml` - Set up `pytest` + `pre-commit` (black, ruff, mypy) - Define Pydantic models: `Disease`, `Gene`, `Match`, `Score`, `Report` - Write ADR-001: Architecture Decision Record (SQLite v1, Pydantic models, CLI via Typer) - **ROGDI data seed**: Hardcode ROGDI facts into test fixtures for validation

Tuesday: Orphanet + OMIM Integration - Implement `disease.py`: query Orphanet API for rare diseases - Filter: monogenic, high-morbidity, no active GT clinical trials - Cross-reference OMIM for gene mapping + inheritance - Cache layer: `requests-cache` for all external APIs - Unit tests with mocked HTTP responses - **ROGDI test**: Verify ORPHA:1946 plus OMIM phenotype 226750 and OMIM gene 614574 resolves correctly

Wednesday: UniProt + AlphaFold Linkage - Implement `homology.py`: map gene → UniProt entry → AlphaFold structure - Fetch protein sequence, domains, GO terms - Store in local SQLite (schema design + `alembic` migration) - Unit tests for protein data pipeline - **ROGDI validation**: Verify ROGDI gene-symbol search → UniProt Q9GZN7, 287 aa, Protein rogdi homolog, PDB 5XQH/5XQI, InterPro IPR028241, Pfam PF10259; explicitly reject obsolete reductase-style scaffold annotations

Thursday: Vector Sizing + Serotype Check - Implement `vector.py`: calculate CDS length, flag AAV packaging limits - Build serotype-to-tissue mapping from literature (static JSON asset v1) - Hard

gate: genes > 4.7kb = auto-lower match score - **ROGDI sizing:** Confirm ~861 bp amino-acid coding region (~864 bp including stop codon) fits comfortably within AAV capacity

Friday: CLI Integration + v0.1 End-to-End - Wire modules into `nanogt match --disease <orphanet_id>` - Single-disease test case: **ROGDI as primary worked example** - Output: JSONL + Markdown report (Jinja2 template) - End-to-end test: from ROGDI disease ID → report in < 2 minutes
- Report must include: gene size, AAV compatibility, cell-type targeting, structural homologs

Week 2: Scoring Shell + ROGDI Deep-Dive

Monday-Tuesday: Scoring Framework - Implement `scoring.py`: tiered scoring (must-pass → weighted → bonus) - Default weights v0.1 (equal across dimensions for baseline) - Manual override system: `--weights-file weights.json` - Top-N ranking with confidence intervals

Wednesday-Thursday: ROGDI Deep-Dive Analysis - Run all 9 scoring dimensions on ROGDI / KTS - Identify closest structural homologs (Foldseek run if available) - Map target cell types: neurons (hippocampus), ameloblasts, renal tubules - Assess delivery routes: IV vs ICV vs intra-oral injection - Evaluate therapeutic window: ROGDI is an intracellular, non-secreted scaffold/adaptor-like protein linked to Rabconnectin-3 / V-ATPase biology — overexpression risk is stoichiometry/mislocalization/complex-disruption risk, not metabolic-catalysis toxicity - Immunogenicity: IEDB epitope scan on ROGDI protein sequence - Codon optimization: CAI for human neurons + ameloblasts - Generate first-pass “Standardised Gene Therapy Protocol” for ROGDI - Document: recommended serotype, promoter, RoA, risk mitigations

Friday: v0.1 Integration Test - Run full pipeline on **ROGDI + 4 additional hand-picked diseases** - Validate ROGDI top match quality against literature - Fix blocking bugs, document known gaps - Tag v0.1-PoC release - Supervisor review checkpoint: ROGDI report

M2: Multi-Disease Batch (Week 3-4)

- Parallelize batch queries with `asyncio` or `concurrent.futures`
- Implement Orphanet full-corpus query (not hand-picked)
- Add `nanogt batch --input diseases.csv --output matches.jsonl`
- Progress bars (`rich`, `tqdm`)
- SQL query layer for post-hoc filtering (`nanogt query --score-min 0.7`)

M3: Full 9-Dimension Scoring (Week 5-6)

- **Immunogenicity:** IEDB API integration, epitope prediction, risk flagging
- **Promoter Matching:** TiProD / EPD / GEO promoter query, enhancer matching
- **Codon Optimization:** CAI scoring relative to tissue (human codon usage)
- **Therapeutic Window:** Overexpression toxicity flags from literature
- **Cross-Correction:** Secreted vs intracellular, rescue radius calculation
- **Weight Tuning:** Run sensitivity analysis, expose `--calibrate` mode

M4: Regulatory Brief Generator (Week 7-8)

- Jinja2 templates for regulatory precedent brief
- Sections: vector precedent, promoter precedent, RoA precedent, CMC analogy

- Auto-populate from matched surrogate data
 - Reference FDA CBER guidance docs + EMA CAT qualification procedures
 - Output: Word compatible Markdown (Pandoc-ready)
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M5: Validation (Week 9-10)

- Retrospective case study: Identify a known successful GT repurposing from literature
 - Run the v0.3 framework on the original disease — did it surface the correct surrogate?
 - If yes: document + publish as proof-of-concept
 - If no: investigate scoring failures, iterate weights, document lessons
 - Second case study (if time): a known failure — did the framework correctly flag the risk?
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M6: Release (Week 11-12)

- GitHub public repo with MIT license
 - Full README with installation, examples, methodology
 - Documentation site (MkDocs or GitHub Pages)
 - Preprint: bioRxiv or medRxiv submission
 - Outreach: patient advocacy orgs, GT biotech BD teams, academic labs
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Risk Register

Risk	Mitigation	Owner	Week to Resolve
Orphanet API rate limits / changes	Cache aggressively; build fallback to OMIM direct	Week 1	
AlphaFold structures missing for rare proteins	Domain-level fallback; validate against experimental structures	Week 2	
No retrospective validation case found	Expand to “platform precedent” not just “approved GT”	Week 9	
Regulatory brief rejected by consulting lawyer	Frame as “informational synthesis” not “legal advice”; include disclaimer	Week 7	

DECISION: Approve plan with expansions

- **Scope:** Approved as SELECTIVE EXPANSION (platform depth + regulatory brief auto-generation)

- **Architecture:** Approved — modular Python CLI, SQLite v1, FastAPI v2
- **DX:** Approved — TTHW target 3.5 min
- **Next step:** /ship when v0.1 PoC is ready (end of Week 2)